

Data 201 : Data Communication & Ethics

Project: Evaluation Dashboard Report for Service Learning Experience at Parami

University on Spring 2025

Professor: Aye Hnin Khine

Student: Nyein Min Soe

A. Introduction

Firstly, the primary reason for doing this work is to show the work I have done during my service learning at Parami university last year. I have wanted to show my personal development during that time, and my enjoyment and self evaluation with the interactive dashboard.

My service learning internship journey begins at the End of July 2025 and lasts till the end of the year. During those times, I have been able to work on different aspects such as learning new skills by attending new courses, improving academic writing, and researching paper developments. This is mainly due to the opportunities given by Parami, and the mentors and people that I have met there. This paper tries to explain more details of the dashboard that I have made to explain the works.

B. Dataset

The dataset is created by the information get from the important aspects of data record from the internship, it also includes all of the tasks, start date, end date and, I have also added the data for personal enjoyment, stress level and the the satisfaction level, using my self evaluation for each of the project, and I have added them by giving them 1 to 10 rating value. Then the data that includes the information of the start date, end date and personal evaluation parameters, to evaluate how my interest and workload has changed over time. Overall I believe it will capture my personal behaviour and subjective and objective aspects of the evaluation matrix during the internship period.

C. Methods

Before creating the dashboard, I have done the exploratory data analysis on the dataset first by getting the summary and the info of the columns. And then I try to format all of the date columns to date time and try to calculate the duration between each column so that I can get the duration of each task. And then I have used the acquired information of date time, and other evaluation matrix columns to get the sum, mean, etc. That data is used to generate the columns for comparing and evaluating the performance of mine during the internship period, and it is reformatted to be able to process with streamlit to get added into the dashboard.

D. Results

Overall, The primary accomplishments are completion of 4 academic courses, reading 2 books, and doing 3 social media posts, during that time. Also, assisted in 3 expert interviews, and made contact with others to understand the relationship.

D(i). Categories and Personal Metrics

Regarding the connection between the Personal Evaluations metrics and Categories, I personally believe that the highest form of enjoyment doesn't necessarily depend on how much I have done but what I am doing for the process and do I feel satisfied or not.

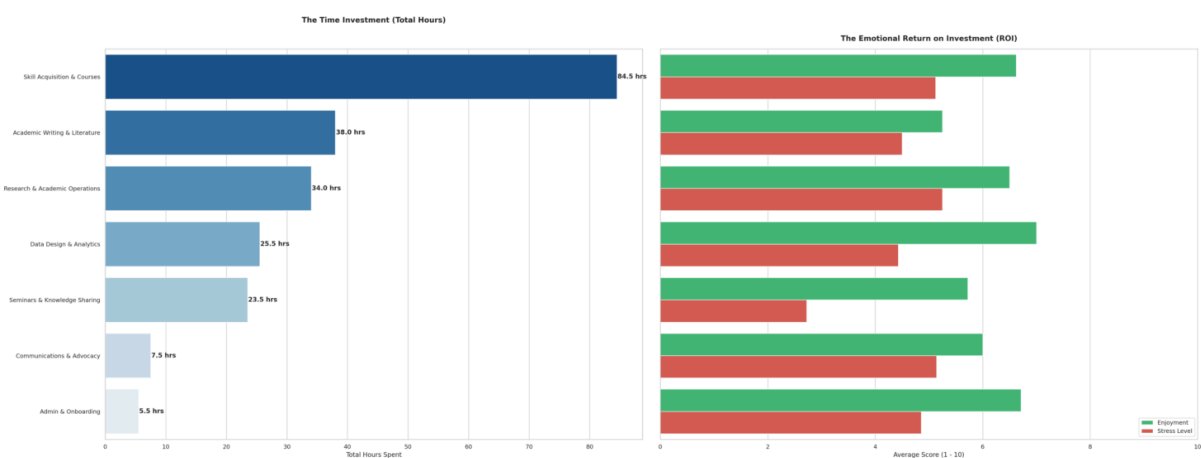


Fig 1 - The Connection between the time and stress level

As seen in figure 1, it is found that the stress level is somewhat lower compared to the engagement, but it does not have any correlation with it. The most heavy workloads are learning new skills, and academic and literature related work. My key take away was that it is essential to do more work on what you enjoy and this will allow you to cope with the work.

To understand more about the correlation between those variables, the correlation matrix is used to determine the correlation between time consumption, stress, enjoyment, and satisfaction level.

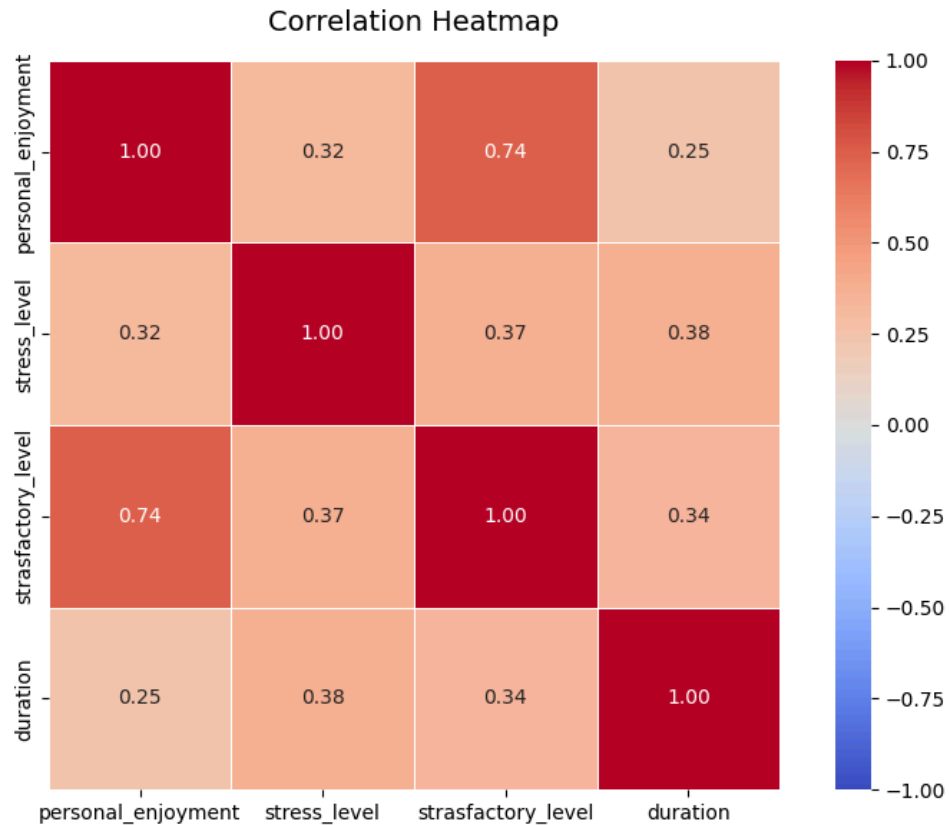


Figure 2 - Correlation Heatmap between Personal Enjoyment, Stress and Satisfactory Level, and Duration of Each Task

The correlation heatmap shows that there is a positive correlation between duration, satisfaction, stress, and personal enjoyment. And satisfaction and personal enjoyment have the strongest correlation as they are strongly representing the same aspect of satisfaction. Secondly, it is important to note that higher working hours does not necessarily have negative effects for this case.

D(ii). Evaluation Metrics Through Time

By following the daily trend of the total working hours through 180 + days of average Stress, and Enjoyment of life, we can see that the working time tends to fluctuate a little bit in the early days of August, days after the September, early October and December, which are most relatively busy weeks for the work. Because the early days of August were the day before the semester begins, and we have to work a lot for that. Secondly, the work pattern follows low workload weeks and high workload weeks alternatively which creates the patterns shown in the fig - 3.

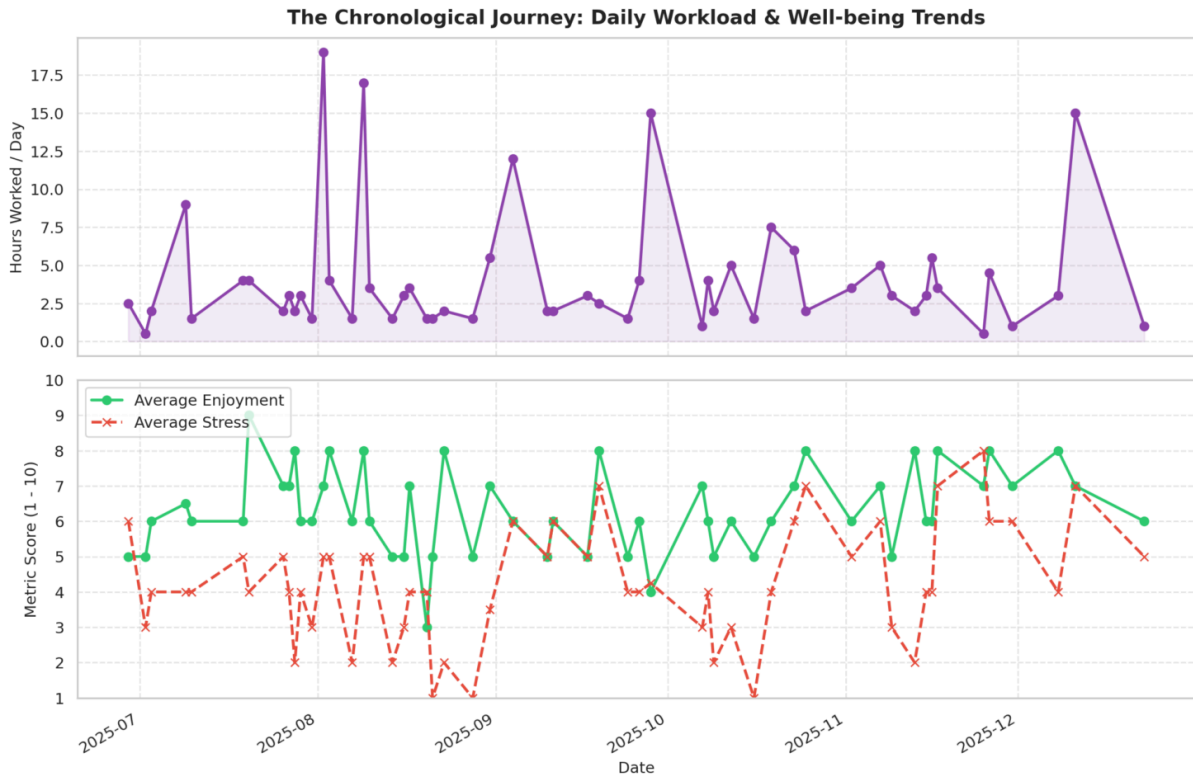


Fig 3 - The Fluctuation of Working hours, Average Enjoyment and Stress Level through Time (Days)

For the enjoyment of the relationship is that the later end of Sep, the middle periods of October, and November are least stressful times. I personally feel that the stress and the conditions and working hours have a bit of connection as sometimes longer working hours cause burnouts, but it is more of a separate issue than the direct relationship.

To understand the overall trend of the metrics, we need less noise plot, which can be obtained through the plot of weeks. Looking at figure 4, we can see that more heavy work has been done during week 4, 5 and 6. That time is during summerbreak, and the most contributions come during that time.

For enjoyment the relationship is stress and enjoyment rhymes, but enjoyment is at a higher pace than stress, and stress fluctuates.

I can see that I personally felt enjoyable overall, but the stress comes during busy weeks, but those weeks are what is worth learning. In figure three we can see that the contents are still described, but it dramatically reduces noise of the plot. Therefore it is given options to choose between the week and days to see on the dashboard for both options of details and readability. It is important to note that the week is predefined with the specific week column counting

from Everyday Thursday of the week (research meetup begins) and the week day order is not skipped even though there were no weeks of working.

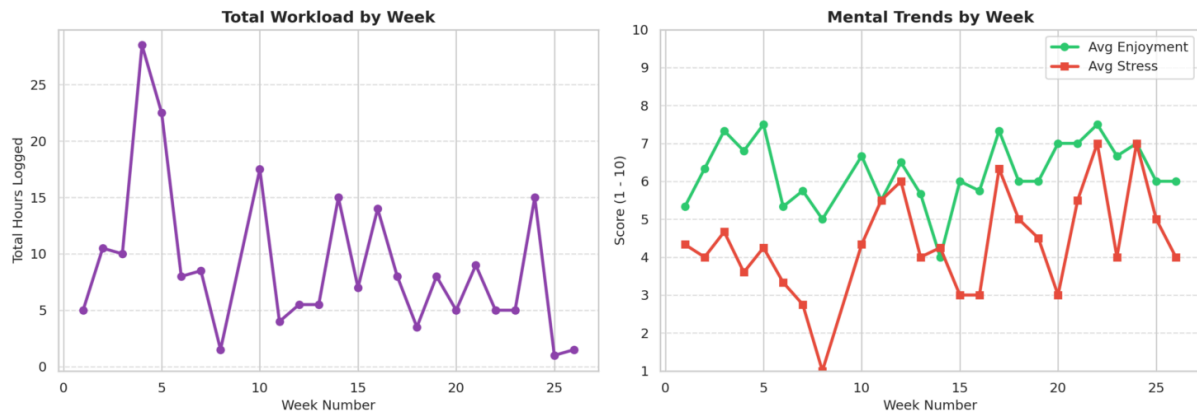


Figure 4 - Trend of Working Hours, Average Enjoyment and Average Stress through each Week

E. Ethical Considerations

Regarding the ethical considerations, I have tried to remove the necessary information file that has been categorized into categories, and all the information including the personal, organizational information is removed, for the anonymity of the data.

F. Conclusion

Overall, the files that are included in the dataset aim to capture more of the information of the personal enjoyment metrics, and the amount of time spent during each task. I have been able to visualize that there is not much of a relationship between the time, and we can see that for the case of my personal story it is far more important for me that more enjoyment encourages me to work on more time, it comes with higher stress levels. Longer durations does not necessarily mean that we have reduced satisfaction for this case. Ethical Considerations for this page includes leaving out sensitive info of the dataset.